DSM Question Bank 4

1) Draw the circuit diagram of SR latch using NOR gates and give its function table.
2) Draw the circuit diagram of SR latch using NAND gates and give its function table.
3) What is the function of the enable in an SR latch? Give the logic diagram of SR latch with control input and its function table.
4) Draw the circuit diagram of D latch and give its function table.
5) How is D latch different from SR latch? What state in SR latch is eliminated in D latch?

The D latch given in Fig. 1 is constructed with four NAND gates and an inverter. Consider the following three other ways for obtaining a D latch. In each case, draw the logic diagram and verify the circuit operation:

6) Use NOR gates for the SR latch part and AND gates for the other two. An inverter may be needed.
7) Use NOR gates for all four gates. Inverters may be needed.
8) Use four NAND gates only (without an inverter). This can be done by connecting the output of the upper gate in Fig. 1 (the gate that goes to the SR latch) to the input of the lower gate (instead of the inverter output).

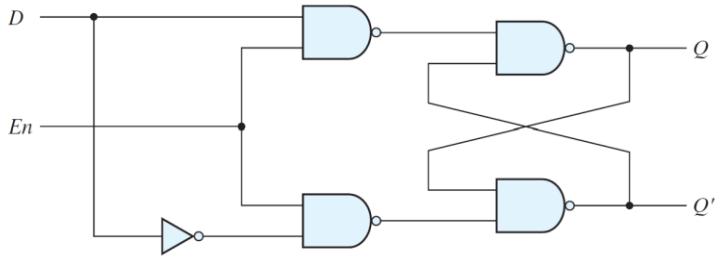
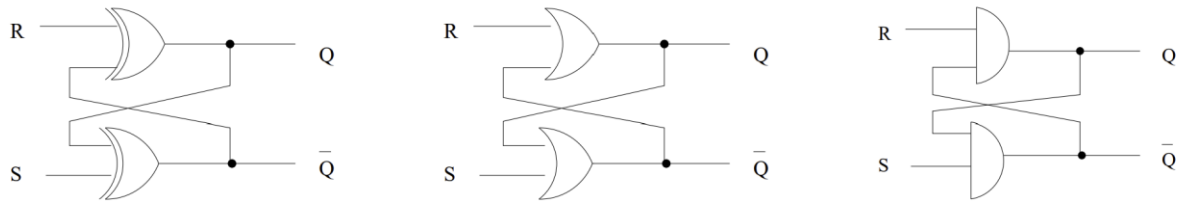


Fig. 1

9) Consider the three circuits below, defined by XOR, OR, AND, NAND gates. In each case, argue whether or not the circuit has essentially the same behaviour as the SR latch (made of NOR gates), namely whether you can use it to hold, set, and reset Q:



10) Consider two types of SR latches: (a) an SR latch made out of NOR gates (b) an SR latch made out of NAND gates. For each type of latch, give an example of R and S timing diagrams that could produce the following timing diagram for Q:



Use a “hold” signal to hold the values after Q changes.